Manav Madan Rawal

Date of Birth: 2nd January 2002
Nationality: Indian
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Education

University of Glasgow 2024 – 2025
Glasgow, Scotland, United Kingdom
Master of Science in Theoretical Physics *14.47/22*

St. Joseph's College (Autonomous), Bengaluru City University 2020 – 2023
Bengaluru, Karnataka, India
Bachelor of Science in Physics, Chemistry, Mathematics *78.78%*

Projects

QCD Internship September, 2025 – Present
Title: Generative AI in Lattice QCD

- Under the guidance of Dr. Chris Bouchard at the University of Glasgow.
- Investigating the application of Generative AI models, such as Normalising Flows, as a high-efficiency alternative to traditional Markov Chain Monte Carlo (MCMC) methods for generating Lattice QCD configurations.
- Developing and training a custom deep generative model in Python to learn the complex probability distribution of lattice gauge fields, with the objective of enabling rapid, parallel sampling.
- Evaluating the performance and physical correctness of AI-generated configurations against those from MCMC, aiming to reduce computational costs and overcome challenges like critical slowing down.

MSc Research Project May, 2025 – September, 2025
Title: QCD colour-flow in top-quark decays [view pdf]

- Under the guidance of Prof. Andy Buckley at the University of Glasgow.
- Investigated the use of QCD colour flow to distinguish colour-singlet hadronic decays, such as $H \to bb$, from their dominant backgrounds.
- Presented a phenomenological comparison between the established Jet Pull observable and the novel Jet Colour Ring (JCR), a theoretically optimal discriminant derived from the likelihood ratio of matrix elements.

Undergraduate Term Paper August, 2022 – October, 2022
Title: Algorithmic Game Theory [view pdf]

- Under the guidance of Prof. Taral D Shah at St. Joseph's College (Autonomous), Bangalore.
- Described the foundations of game theory, nash equilibria, Braess' paradox and its known applications.
- Possible solutions to real life problems using Braess' Paradox and Game Theory.

Geometric Integration Techniques August, 2023 – October, 2023
Title: 'Folklore Result' - The Geometric Reduction of Integration [view pdf]

- Worked to identify results in Calculus which are not usually taught in courses but greatly aid in computing integrals faster.
- Involves computing an integral of a function over one domain by knowing the value of the integral of the function over another domain.
- Finding solutions in a simpler way using ideas from Geometry.

Test Scores and Results

IELTS 2024

Secured an overall Band score of 8.5

- Listening - 9.0
- Speaking - 8.5
- Reading - 8.5
- Writing - 7.5
- Click here to view certificate

IIT - JAM 2024

Qualified the Indian Institute of Technology (IIT) - Joint Admission test for Masters (JAM) - Physics 2024

PG - CUET 2024

Qualified the Postgraduate Common Univerity Entrance Test (PG-CUET) - Physics 2024

Other Academic Achievements

International Leadership Scholarship 2024

Secured the highly competitive 'University of Glasgow International Leadership Scholarship'.

SOF - IEO

Secured International Rank 36 in Round 1 of the International English Olympiad (IEO) conducted by the Science Olympiad Foundation (SOF) in High School.

SAGE 2016

Secured All India Rank 1 in the Social Science Olympiad conducted by the SAGE Foundation

Extracurricular Activities

Community Outreach 2021-2023

St. Joseph's College (Autonomous)

- Rural exposure, field visits, helping underprivileged children in Solur, Karnataka
- 'Save Water' drive in downtown Bengaluru
- Click here to view certificate

Computer and Communication Skills

Programming Languages: Python, C, C++, R, Julia, Java, HTML-CSS, JavaScript, LaTeX, Git, Docker, Bash
Click here to view Github page
Particle Physics Tools: RIVET, Pythia
Database: MySQL, NoSQL
MS Office
Languages: English (Native/Bilingual Proficiency), Hindi (Native/Bilingual proficiency), Gujarati (Native/Bilingual proficiency), Kannada (Limited Working Proficiency)